

IoT Deliverable 3

AC/DCI Group

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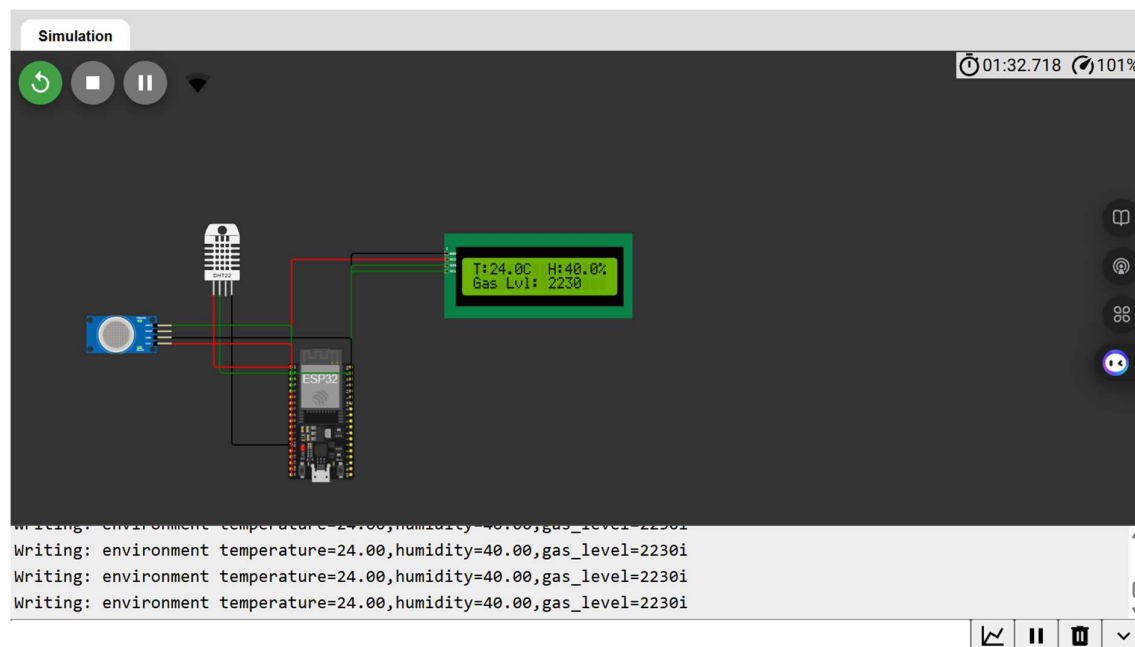
1. The Simulation

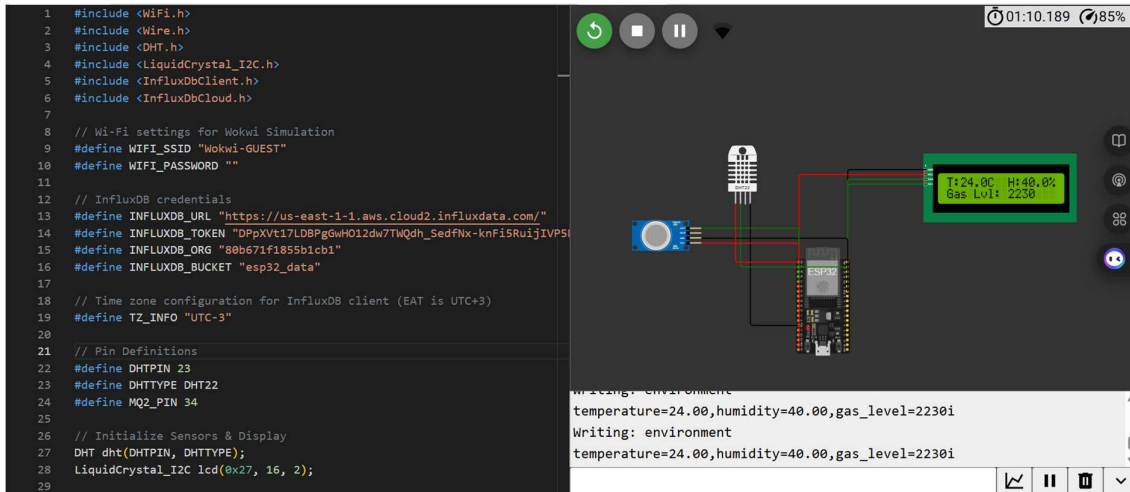
Architecture: 1 ESP32 connected to 1 MQ-2 (Gas Sensor), 1 DHT22 (Temperature & Humidity), and 1 LCD (I2C).

This architecture serves as the edge device for our cloud integration. The ESP32 is programmed to sample environmental metrics: temperature, humidity, and gas concentration every 10 seconds. It displays these variables locally on the LCD screen while simultaneously establishing a Wi-Fi connection to transmit the data payload to the cloud via an HTTP API. The simulation was successfully compiled and executed online using the Wokwi web environment.

Simulated Model Link: <https://wokwi.com/projects/469158779733863425>

Simulation Execution:





2. InfluxDB (Cloud Data Storage)

InfluxDB Cloud acts as the time-series database for this project. Once connected to Wi-Fi, the ESP32 utilizes a secure API token, Organization ID, and URL to push data into our designated cloud bucket (esp32_data). The telemetry data is structured using Line Protocol and grouped under an "environment" measurement, allowing the database to efficiently timestamp and store the temperature, humidity, and gas level data streams for querying.

Data Storage Evidence:

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Data Explorer

+ New Script OPEN SAVE

Schema Browser SQL Sync

Bucket: esp32_data

Measurement: environment

Search fields and tag keys

Fields: gas_level, humidity, temperature

Tag Keys: No Tags Found

Query: `SELECT * FROM "environment" WHERE time >= now() - interval '1 hour'`

Ready (312ms) CSV Past 15m RUN

Search results... 83 rows TABLE GRAPH

gas_level	humidity	temperature	time
no group long	no group double	no group double	no group dateTime:RFC3339
3628	40	24	2026-07-10T18:35:50.911Z
3628	40	24	2026-07-10T18:36:19.611Z

1 2 3 4 5 ... 28

3. Grafana (Data Visualization)

Grafana is integrated directly with our InfluxDB cloud instance to provide real-time, industry-standard visual analytics. Using the Flux scripting language to query the database, we configured a dashboard with three distinct panels to monitor the physical environment:

- **Live Temperature:** A stat panel displaying the most recent temperature reading in Celsius.
- **Current Gas Level:** A gauge panel configured with color-coded health thresholds (Green for normal, Yellow for warning, Red for critical) to monitor the analog MQ-2 readings.
- **Environmental Metrics:** A time-series graph that tracks historical temperature and humidity trends.

Grafana Dashboard Link: <https://boldflax650.grafana.net/goto/s54jlj?orgId=stacks-1719195>

ESP32 Sensor Dashboard:



4. Groupwork Evidence

The following evidence verifies the active participation of all group members during the circuit design, cloud configuration, and dashboard integration phases of this deliverable.

